Unit 6, Lesson 12

Solve Problems Involving Multiplication

Standards

Addressing	**4.MD.A.2, 4.NBT.B.5**
Building Toward	**4.MD.A.2**

Instructional Routines

- MLR7 Compare and Connect
- What Do You Know about ____?

Goals

- Explain (orally and in writing) strategies for solving problems that involve multiplication of multi-digit numbers.

Student Facing Learning Goals

Let's solve problems using what we learned about multiplication of whole numbers.

Lesson Purpose

The purpose of this lesson is for students to solve contextual problems that involve multiplication of a single-digit number and a whole number of up to four digits, or multiplication of 2 two-digit numbers.

Narrative

This lesson gives students the opportunity to apply the multiplication strategies they have learned to solve various contextual problems involving measurement. The problems vary in format and complexity—some involve a single computation and others require multiple steps to solve. The work here prompts students to make sense of problems and persevere in solving them (MP1) and to reason quantitatively and abstractly (MP2).

Access for Students with Disabilities

- Action and Expression

Required Materials

Materials to Gather

- Tools for creating a display: Activity 1

Lesson Timeline

Warm-up	10 min
Activity 1	25 min
Activity 2	10 min
Synthesis Estimate	10 min
Cool-down	5 min

Teacher Reflection Questions

As students shared their ideas today, how did you ensure all students' voices were heard and valued as an important part of the collective learning?

Warm-up

10 min

What Do You Know about 1 Year?

Standards

Building Toward **4.MD.A.2**

Instructional Routines

- What Do You Know about _____?

The purpose of this *Warm-up* is to elicit students' knowledge about 1 year as a measure of time and the ways it can be represented. The reasoning and conversations here will be helpful as students solve problems that involve time in years later in the lesson.

Student Task Statement

What do you know about 1 year?

Student Response

Sample responses:

- It is 365 days long.
- It is 12 months long.
- Sometimes 1 year has an extra day.
- It is the time it takes the earth to make a rotation around the sun.

Launch

- Display: "1 year"
- "What do you know about 1 year?"
- 1 minute: quiet think time

Activity

- Record responses.

Activity Synthesis

- "What does a year measure?" (time)
- "What are some aspects of time that are related to a year?" (seasons, months, weeks, days)

Activity 1

25 min

Time Flies When We Leap Years

Standards

Addressing **4.MD.A.2, 4.NBT.B.5**

Instructional Routines

- MLR7 Compare and Connect

In this activity, students use what they learned about multiplication of multi-digit numbers and unit conversion to solve problems involving measurements. Students may choose to represent the situations in a number of ways—concretely or visually (by drawing diagrams) or abstractly (by writing expressions and equations). While some problems can be reasoned additively, students may opt to reason multiplicatively for practical reasons.

Regardless of their chosen representations and reasoning strategy, students reason quantitatively and abstractly when they interpret and solve the questions about different units of time (MP2).

This activity uses *MLR7 Compare and Connect*. Advances: representing, conversing

Access for English Language Learners

Action and Expression: Develop Expression and Communication. Provide access to a variety of tools, including colored pencils, grid paper, base-ten blocks, and a visual display that students can use as a reference showing a variety of strategies from throughout the section.
Supports accessibility for: Conceptual Processing, Organization, Memory

Required Materials

Materials to Gather

- Tools for creating a display: Activity 1

Student Task Statement

1. A baby elephant was born exactly 48 weeks ago. How many days old is the elephant?
2. A leap year has 366 days. A non-leap year has 365 days. How many days are in 3 leap years?
3. In our calendar system, some months are 31 days long, some are 30 days long, and one month (February) is either 28 or 29 days long.

   What if the calendar system changed so that each month has 31 days? How many more days would there be in a year?



Student Response

1. $48 \times 7 = 336$
2. $366 \times 3 = 1{,}098$
3. 7 days more than a common year (or 6 days more than a leap year). Sample responses:
   - $12 \times 31 = 372$ and 372 is 7 more than 365 and 6 more than 366.
   - In a common year, there are four months that are 30 days long and one that is 28 days long. To make these five months 31 days each means adding $(4 \times 1) + 3$ or 7 days. In a leap year, it'd mean adding $(4 \times 1) + 2$ or 6 days.

Launch

- Groups of 2–4
- Give each group tools for creating a visual display.
- "Today we'll solve some problems that involve measurements."
- "Before we solve any problem, let's do some estimation."
- Read the first problem as a class.
- "Estimate: About how many days old is the baby elephant?"
- 1 minute: quiet think time
- Display the following ranges and poll the class on their estimate. (Alternatively, post each range in a different part of the classroom and ask students to stand by the range that reflects their estimate.)
  - 280 to 299
  - 300 to 349
  - 350 to 400
- Select a student who selected each range to explain their estimate. Then ask students if they'd revise their estimate.

Activity

- "Take a few quiet minutes to solve the first two problems. Then, discuss your responses with your group and work on the last problem together."
- 5 minutes: independent work time
- 5 minutes: group work time
- Monitor for the different representations and strategies used to solve the problems.
- Assign one problem to each group.

MLR7 Compare and Connect

- "Create a visual display to show how you solved the problem assigned to your group. Organize your work so that it can be followed by others."
- 3 minutes: group work time
- 5 minutes: gallery walk
- Prompt students to make note of the different strategies they see and any solutions that they question.

Activity Synthesis

- "Most of the problems can be solved by multiplication. What is the same between the solutions and multiplication strategies that you saw? What's different?"
- 30 seconds quiet think time
- 1 minute: partner discussion
- Record students' responses, or display students' diagrams and representations.
- "What are some common strategies for multiplying a multi-digit number by a one-digit number (48 and 7, or 366 and 3)?"
- "What are some strategies you saw for multiplying 2 two-digit numbers (12 and 31)?"

Activity 2

10 min

Coin Collection

Standards

Addressing **4.MD.A.2, 4.NBT.B.5**

This activity offers students more practice with using multiplication to solve contextual problems (MP2), including situations in which at least one factor is four digits long, and to generate a new problem according to some parameters.

Student Task Statement

1. Lin's family collects 2,074 nickels. How many pennies are worth the same amount?
2. If Lin's family saves 2,074 nickels each year for 4 years, how many nickels will her family have?
3. Create a situation that involves a problem that

Launch

- Groups of 2–4

Activity

- 6–7 minutes: independent work time

can be solved by finding the value of $8 \times 1{,}049$. Solve the problem. Explain or show your reasoning.

Student Response

1. $2{,}074 \times 5 = 10{,}370$
2. $2{,}074 \times 4 = 8{,}296$ (1 fewer group of 2,074 than in 10,370, or $10{,}370 - 2{,}074$)
3. Sample response: The local library had 1,049 books checked out each day for 8 days. How many books were checked out in total?
   $8 \times 1{,}000 + 8 \times 40 + 4 \times 9 = 8{,}392$

- 2–3 minutes: group discussion
- Monitor for and select different ways students represented the situations and solved the problems.

Activity Synthesis

- "In the *Lesson Synthesis*, let's compare some different strategies."

Lesson Synthesis

"Today we used what we have learned about multiplication to solve problems involving measurement."

Invite selected students to share their responses to the problems in the last activity. As each student shares, ask if others in the class solved it the same way and if they approached it differently.

Prompt students to explain what their numerical solutions represent in each situation.

"How would you know if your solutions were correct?" (I used another strategy to see if I got the same answer. I estimated first so that I had an idea how big or small the answer would be. I checked with my groupmates.)

Consider asking: "When you had to multiply numbers, which method did you rely on the most? What made you choose that method?"

Suggested Centers

- Five in a Row: Multiplication (3–5), Stage 3: Two-Digit Factors (Addressing)
- Compare (1–5), Stage 3: Multiply within 100 (Supporting)

Cool-down

5 min

Leap Year

Standards

Addressing **4.MD.A.2, 4.NBT.B.5**

Student Task Statement

In a leap year, the month of February has 29 days. How many hours are in that month? Explain or show your reasoning.

Student Response

696 hours. Sample response:

$$\begin{array}{r} 29 \\ \times\ 24 \\ \hline 36 \\ 80 \\ 180 \\ +\ 400 \\ \hline 696 \end{array}$$

Responding to Student Thinking

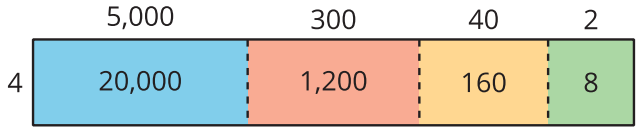
Students who use an algorithm that uses partial-products may neglect to consider the place value of the digits they are multiplying and end up with some incorrect partial products.

Next Day Supports
Before the *Warm-up*, select a student's *Cool-down* from the previous lesson (name anonymous). Ask students to identify what the student did well and what the student needs to do to improve the cool-down.

Section B Summary

We learned to multiply factors whose products are greater than 100, using different representations and strategies.

When working with multi-digit factors, it helps to decompose them by place value before multiplying. For example, to find the value of $4 \times 5{,}342$, we can decompose the 5,342 into its expanded form, $5{,}000 + 300 + 40 + 2$, and then use a diagram or an algorithm to help us multiply.



20,000 + 1,200 + 160 + 8 = 21,368

$$\begin{array}{r} 5,342 \\ \times\ \ \ 4 \\ \hline 8 \\ 160 \\ 1,200 \\ +\ 20,000 \\ \hline 21,368 \end{array}$$

In both the diagram and the algorithm, the 20,000, 1,200, 160, and 8 are called the partial products. They are the result of multiplying each decomposed part of 5,342 by 4.

We can do the same to multiply a two-digit number by another two-digit number.

For example, here are two ways to find the value of 31×15. The 31 is decomposed into $30 + 1$ and 15 is decomposed into $10 + 5$.

$$\begin{array}{r} 31 \\ \times\ 15 \\ \hline 5 \\ 150 \\ 10 \\ +\ 300 \\ \hline 465 \end{array}$$

